

What is the length of an array?

The **length (or size)** of an array means:

How many elements are stored in the array

Example:

```
int marks[5] = {60, 70, 80, 90, 100};
```

→ The length of this array is **5** because it has **5 values**.

Method 1: Finding array length using `sizeof` (Most Common in C++)

Formula

```
array_length = sizeof(array) / sizeof(array[0]);
```

Why this works

- `sizeof(array)` → total memory used by the array
 - `sizeof(array[0])` → memory used by **one element**
 - Dividing gives the **number of elements**
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Example 1: Integer Array

```
#include <iostream>
using namespace std;

int main() {
    int numbers[] = {10, 20, 30, 40, 50};

    int length = sizeof(numbers) / sizeof(numbers[0]);

    cout << "Length of array: " << length;

    return 0;
}
```

Output

```
Length of array: 5
```

Example 2: Character Array

```
#include <iostream>
using namespace std;

int main() {
    char vowels[] = {'a', 'e', 'i', 'o', 'u'};

    int length = sizeof(vowels) / sizeof(vowels[0]);

    cout << "Number of vowels: " << length;

    return 0;
}
```

Output

```
Number of vowels: 5
```

Method 2: Length of a String (Character Array)

Using `strlen()` (for strings only)

```
#include <iostream>
#include <cstring>
using namespace std;

int main() {
    char name[] = "Yasir";

    cout << "Length of name: " << strlen(name);

    return 0;
}
```

Output

```
Length of name: 5
```

◆ `strlen()` counts characters **excluding** the `'\0'` (null character).

Method 3: Using Loop (Manual Counting – Learning Purpose)

```
#include <iostream>
using namespace std;

int main() {
    int arr[] = {2, 4, 6, 8, 10};
    int count = 0;

    for(int i = 0; i < 5; i++) {
        count++;
    }

    cout << "Array length: " << count;

    return 0;
}
```

⚠ This works only when you **already know the size**.

Important Note ⚠

✗ You **cannot** find the length of an array using `sizeof` inside a function when passed as a parameter.

Example (Wrong):

```
void show(int arr[]) {
    cout << sizeof(arr);    // gives size of pointer, not array
}
```

✓ Correct way: pass the size separately.

Summary Table

Method	Used For	Beginner Friendly
<code>sizeof(array)/sizeof(array[0])</code>	Any array	✓ Best
<code>strlen()</code>	Strings only	✓
Loop counting	Practice only	⚠

Related Topics

- [C++ One-Dimensional Arrays — Syntax & Examples](#)
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